(v) $P(0) = 3C_0\ 0.3^0\ (0.7)^{3-0} = 0.7^3 = 0.343$

(vi) $F(0) = N \times P(0) = 100 \times 0.343 = 34.3$

(vii) $F(x+1) = \frac{n-x}{x+1} \times \frac{p}{q} \times F(x)$

$\therefore$ $F(1) = F(0+1) = \frac{3-0}{0+1} \times \frac{0.3}{0.7} \times 34.3 = 44.247$

$F(2) = F(1+1) = \frac{3-1}{1+1} \times \frac{0.3}{0.7} \times 44.247 = 19.03$

$F(3) = F(2+1) = \frac{3-2}{2+1} \times \frac{0.3}{0.7} \times 19.03 = 2.727$

Solution:

(i) The fitted binomial distribution is

$P(X = x) = 3C_x\ 0.3^x\ 0.7^{(3-x)}, \quad x = 0,1,2,3$

(ii) The expected frequencies are :

x	0	1	2	3	Total
Observed frequencies (O_i)	36	40	22	2	100
Expected Frequencies (E_i)	34	44	19	3	100

10.3.2 Fitting of Poisson Distribution

When a Poisson distribution is to be fitted to an observed data the following procedure is adopted:

(i) Find the mean: $\bar{x} = \frac{\Sigma fx}{\Sigma f}$

(ii) Poisson parameter = $\lambda = \bar{x}$

(iii) Probability mass function is: $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ $x = 0, 1, 2\ldots.$

(iv) Put X = 0 and find $P(0) = \frac{e^{-\lambda}\lambda^0}{0!} = e^{-\lambda}$

(v) Expected frequency for $x = 0$ is $F(0) = N \times P(0)$, $\Sigma f_i = N$

(vi) Other expected frequencies can be found using

$F(x+1) = \frac{\lambda}{x+1} \times F(x)$ for $x = 0, 1, 2\ldots.$

Example 10.35

The following mistakes per page were observed in a book

Number of Mistakes (per page)	0	1	2	3	4
Number of pages	211	90	19	5	0

Fit a Poisson distribution and estimate the expected frequencies.